

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2023****Course: B. Tech.****Branch : Computer Engineering****Semester: V****Subject Code & Name: Machine Learning [BTCOC503]****Max. Marks: 60****Date: 11/08/2023****Duration:03:00 Hrs.****Instructions to the Students:**

- 1 All the questions are compulsory.
- 2 The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- 3 Use of non-programmable scientific calculators is allowed.
- 4 Assume suitable data wherever necessary and mention it clearly.

(Level /
CO) Marks**Q. 1 Solve Any Two of the following.****12**

a) Explain Linear Regression algorithm.

b) Explain following terms with respect to linear regression:

A)

1. Cost function

1

2. Gradient Decent

B)

Explain different splitting techniques of decision tree. Explain the basic terminologies of decision tree.

1**C)**

What do you mean of under fitting? What are the different reasons of under fitting? How under fitting in model can be reduced.

1**Q. 2 Solve Any Two of the following.****12****A)** Solve the following problem to find out whether car is stolen or not using the Naive Baye's algorithm.

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

2**B)** Explain the features and types of Logistic Regression.**2****C)** Explain the working of the Support Vector Machine algorithm.**2**

Q. 3 Solve Any Two of the following.	12
A) Explain the working of Perceptron with suitable diagram.	4
B) What is backpropagation? Explain in brief. Also, discussed what are the needs of backpropagation.	4
C) Explain working of the multilayer network.	4
Q. 4 Solve Any Two of the following.	12
A) What is ensemble learning? Describe in detail the types of ensemble learning.	3
B) Explain any one ensemble learning algorithm.	3
C) Write note on Computational Learning Theory.	3
Q. 5 Solve Any Two of the following.	12
A) Explain the how K-Means Clustering Algorithm works? Explain the elbow method.	2
B) What is unsupervised machine learning? Explain with suitable example. What is clustering? https://www.batuonline.com	1
C) Explain Hierarchical Clustering Technique and its types in detail.	2
*** End ***	